

EMPOWERING VISUALLY IMPAIRED INDIVIDUALS WITH HOLISTIC
ASSISTANCE USING REAL-TIME SPATIAL AWARENESS SYSTEM

BY
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AUTHORSHIP STATEMENT

I, Xinrui Yu (X. Y.), attest that the work in this thesis is substantially my own.

In accordance with the disciplinary norm of Electrical Engineering (see IIT Faculty Handbook, Appendix S), the following collaborations occurred in the thesis:

Dr. Jafar Saniie (J. S.) of the Illinois Institute of Technology provided scientific guidance and advice on computer vision based assistance and human-machine interface for the visually impaired.

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ABSTRACT

The integration of artificial intelligence (AI) into daily life opens unprecedented avenues for enhancing the experiences of visually impaired individuals, offering them greater autonomy and quality of life. This thesis introduces a Visually Impaired Spatial Awareness (VISA) system designed to assist visually impaired individuals holistically through a structured approach. At the foundational level, the VISA system incorporates several key technologies to interpret the surroundings and assist in basic navigation tasks. It utilizes Augmented Reality (AR) markers to facilitate recognition of places and aid in navigation, employs neural network models for advanced object detection and tracking, and leverages depth information for accurate object localization. Progressing to the intermediate level, the VISA system integrates the data obtained from object detection and depth sensing to assist in more complex navigational tasks such as obstacle avoidance and pathfinding toward a desired destination. At the advanced level, the VISA system synthesizes the capabilities developed at the foundational and intermediate levels to enhance the spatial awareness of visually impaired users, allowing them to undertake complex tasks, such as navigating complex environments and locating specific items. The VISA system also emphasizes efficient human-machine interaction, incorporating text-to-speech and speech-to-text technologies to facilitate natural and intuitive communication between the user and the system. The VISA system's performance was evaluated in different environments simulating real-world scenarios. The experimental results show that the user can interact with our system intuitively with minimal effort, and affirm that the VISA system can effectively assist the visually impaired user in locating and reaching for objects, navigating indoors, identifying merchandise, and recognizing both handwritten and printed texts.

CHAPTER 1

INTRODUCTION

According to the Global Vision Database 2019 Blindness and Vision Impairment Collaborators, the year 2020 saw approximately 43.3 million people living with blindness, and another 295 million people experience moderate vision impairments and above. Projections suggest a significant increase by 2050, with the blind population expected to rise to 61.0 million, and those with moderate to severe vision impairments expanding to 474 million individuals [1]. In the United States alone, there are more than one million blind people in the year 2015, and that number is projected to double in the year 2050 [2]. These statistics highlight an escalating global health concern that necessitates immediate attention and action. There can be no overstatement about the importance of vision. It is a fundamental sensory modality that underpins a myriad of daily activities, including but not limited to navigation, fetching objects, reading, and engaging in other complex tasks, all of which are integral to personal independence and quality of life [3][4][5].

Of all the tasks involving vision, there is no single most important task above all others; many tasks share similar importance, and completing one without another cannot be considered satisfactory. Navigating indoor spaces poses a unique set of challenges for visually impaired individuals, often complicating what many would consider routine activities [6]. Identifying and locating everyday objects such as clothes, food, and electronics is essential for performing daily routines and tasks [7]. Similarly, the ability to move safely and efficiently through various indoor environments without encountering obstacles is fundamental to maintaining an independent lifestyle [8]. Moreover, access to information through reading and writing, whether via tactile methods like braille or through assistive digital technologies, is crucial for personal, educational, and professional development [9]. Finally, the capacity for so-

cial interaction and communication plays an important role in fostering relationships, community engagement, and accessing support when needed [10].

Addressing these tasks, both individually and collectively, is central to empowering visually impaired individuals to navigate not only basic but also complex tasks with greater confidence and autonomy. For instance, effectively managing these tasks can significantly enhance the ability of a visually impaired person to perform activities such as grocery shopping, which requires a combination of navigation, object identification, information access, and sometimes, communication with others [11].

Even before the advent of computer vision, many methods, tools, and systems were developed to assist the visually impaired in the tasks listed above. Some common examples include the white cane, guide dogs, and braille. However, without machine vision and AI technologies, these methods were inherently lacking in their effectiveness and fields of application. The white cane, for instance, while invaluable for immediate spatial detection, offers a limited range and no object identification capabilities [12]. Guide dogs, often considered the best alternative to sighted assistance, offer companionship, increased mobility, and sometimes even higher social status [13][14]. Yet, they come with high training and acquisition costs, making them mostly unavailable for low-income individuals [15]. Also, despite their ability to navigate complex environments, guide dogs cannot communicate specific object information to their handlers [16]. Braille has revolutionized access to written information for the visually impaired, but it is confined to touching and needs to be printed beforehand, limiting its capacity to deliver immediate and dynamic content [17].

With the rapid advancements in the field of computer and machine vision, the landscape of assistive technologies for the visually impaired is undergoing a transformation. The advent of Text-to-Speech (TTS) and Speech-to-Text (STT) technologies can greatly improve the interfacing options for the visually impaired [18]. Coupled

with the emergence of deep learning algorithms, tasks such as object recognition, which were once challenging, are now attainable and can be integrated into practical applications. Moreover, the progress in embedded systems and system-on-chip (SoC) technologies heralds the advent of portable and wearable smart devices tailored to the needs of the visually impaired [19]. These devices are designed to assist with a range of activities, notably indoor navigation and environment perception. The promise of a unified system that can seamlessly integrate various functionalities—from navigation assistance to object and text recognition—is within reach. However, many existing systems tend to specialize in singular aspects of assistance [20][21].

To address the deficiencies of the approaches above, this thesis proposes a holistic approach, leveraging the full spectrum of current technologies to create a comprehensive assistive system that transcends the limitations of specialized aids. Our VISA system utilizes cutting-edge AI technologies, including advanced object detection and spatial navigation algorithms, alongside user-friendly interfaces to assist visually impaired users in overcoming the challenges associated with these basic tasks. By focusing on these foundational areas, the proposed VISA system aims to provide a comprehensive solution that can be adapted and expanded to meet a wide range of needs and activities, ultimately facilitating a more accessible and navigable environment for visually impaired individuals.

1.1 Layered Approach to Holistic Assistance for Visually Impaired Users

Navigating indoor environments poses a unique set of challenges for visually impaired individuals, necessitating a comprehensive and holistic approach to assistive technology. Recognizing the diversity of tasks and the varying levels of complexity involved, our VISA system adopts a layered approach to holistic assistance. This methodology is designed to empower the user to understand, interact with, and navigate through indoor spaces safely and independently, as shown in Figure 1.1.

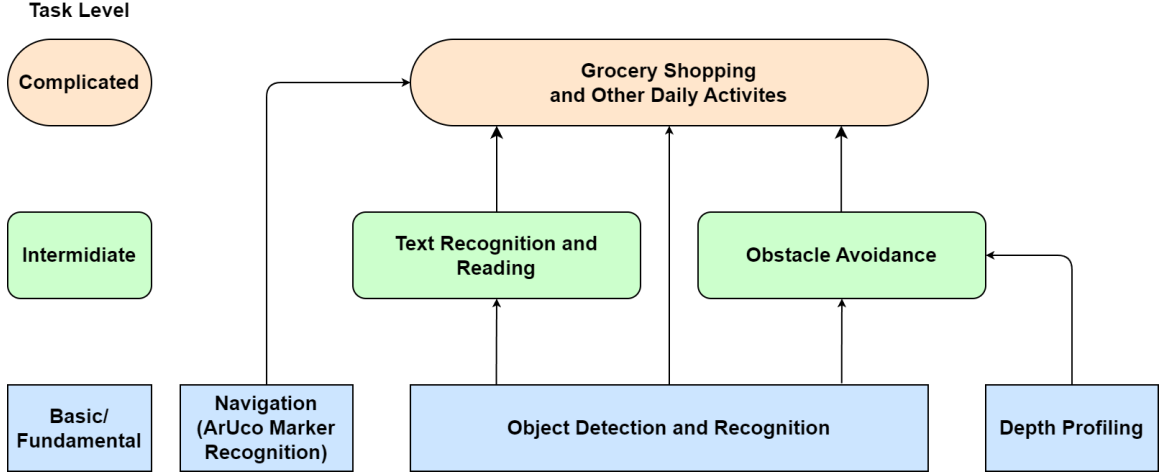


Figure 1.1. Layered Approach for Holistic Assistance

Foundation Level - Establishing Spatial Awareness: At the base of our approach lies the establishment of spatial awareness through the integration of ArUco (Augmented Reality University of Cordoba) markers [22] for navigation, advanced object detection and recognition using YOLOv8 (You Only Look Once, version 8) neural network model [23], and depth sensing technologies. These foundational tasks aim at creating a basic yet essential understanding of the user's immediate surroundings. By identifying ArUco markers for navigation, detecting and tracking objects, and localizing objects in space, we lay the groundwork for more complex tasks. This level is essential for users to gain an initial sense of direction and to understand the layout of the environment around them.

Intermediate Level - Enhancing Navigation and Safety: Building upon the foundational layer, the intermediate level focuses on synthesizing spatial data to facilitate obstacle avoidance and efficient pathfinding toward a desired destination. This layer leverages the spatial awareness established earlier to provide real-time guidance, helping users avoid obstacles and navigate through indoor spaces more confidently. It represents a crucial step in enhancing the independence and safety of

visually impaired individuals as they move within indoor environments.

Advanced Level - Facilitating Complex Interactions: The apex of our approach integrates all the information processed in the foundational and intermediate layers to assist users in performing complex tasks, such as grocery shopping or finding specific items within an indoor space. This advanced layer of tasks is where the VISA system showcases its potential to significantly improve the autonomy of visually impaired users, allowing for a level of interaction with their environment that mimics the navigational capabilities of sighted individuals.

Integration - Human-Machine Interface for Seamless Interaction: Across all layers, our VISA system employs an intuitive human-machine interface, utilizing text-to-speech and speech-to-text technologies. This ensures that users can interact with the VISA system effortlessly, receiving auditory feedback that guides them through their environment, listening to recognized texts, and allowing them to communicate their needs and destinations.

By adopting this layered approach, our VISA system not only addresses the immediate challenges of object recognition and indoor navigation, but also opens up new possibilities for visually impaired individuals to engage with their surroundings in meaningful and empowering ways. We will explore each layer in detail, discussing the specific technologies and methodologies employed and how they contribute to the overall goal of enhancing life quality for visually impaired users.

1.2 Organization of the Thesis

In the following chapters, we will delve into the specific technologies and methodologies employed by the proposed VISA system, discuss its implementation and evaluation, and explore the potential it holds for transforming the everyday experiences of visually impaired users.

Chapter 2 introduces related works in the field of object recognition systems, obstacle avoidance systems, and indoor navigation systems. Chapter 3 introduces an RGB-D camera-based object recognition and localization module, aimed to assist visually impaired users in identifying and locating everyday objects around them. Chapter 4 introduces a navigation module based on ArUco marker recognition and depth information acquired by the RGB-D camera. It enables the visually impaired user to move to a desired location safely while avoiding obstacles. Chapter 5 introduces a friendly human-machine interface utilizing speech-to-text, text-to-speech, and Google Lens, together with a comparison of our VISA system versus some existing systems or items to assist the visually impaired. Chapter 6 explains that by integrating the modules described above, we can assist the visually impaired user in performing complex activities, such as grocery shopping, and shows testing results. Finally, in Chapter 7, we go into the conclusion of the thesis.

CHAPTER 2

LITERATURE REVIEW

In recent years, numerous electronic systems and devices have been developed to assist the visually impaired in different aspects of their lives [20]. In the pursuit of holistic assistance for visually impaired individuals, a thorough examination of existing literature across three key domains—object recognition systems [24][25], obstacle avoidance systems [26][27], and indoor navigation systems [28][29][30][31] —is imperative. These areas are instrumental in the development of our VISA system designed to enhance the autonomy of visually impaired users in indoor environments. Also, we specifically look into the overlapping of technologies for different tasks, for example, if one category of sensors can assist in object recognition, obstacle avoidance, and navigation. Finally, we examine the processing unit used by all the systems and try to analyze the trend of system evolution. The conclusions reached can help us settle on our design of the VISA system.

2.1 Object Recognition Systems

Object recognition has priority in many assistance systems designed for visually impaired individuals. Over the past decade, advances in computer vision and artificial intelligence have paved the way for systems capable of identifying and locating objects in real-time. Of those systems, most leverage deep learning techniques for greater efficiency and accuracy. Currently, research in this field mainly utilizes a number of neural network algorithms, including but not limited to You Only Look Once (YOLO) [32][33][34], Faster R-CNN [35], and Single Shot MultiBox Detector (SSD) [36]. Of those algorithms, YOLOv8 is among the most recent ones and offers superior performance compared with many other more dated algorithms [23].

2.1.1 Object Recognition using YOLO.

The YOLO (You Only Look Once) algorithm represents a cutting-edge achievement in the field of computer vision, particularly in the domain of real-time object detection. Introduced by Redmon et al. in 2016 [32], YOLO revolutionizes deep learning-based object recognition by enabling the processing of images in a single pass instead of processing interesting parts of an image, thus significantly reducing detection time while maintaining high accuracy. This processing procedure is shown in Figure 2.1. Unlike traditional object detection methods that complete the task in two stages—first proposing potential bounding boxes and then classifying those regions—YOLO unifies these steps into one cohesive framework. This integration allows YOLO to predict bounding boxes and class probabilities directly from a full image in one evaluation, making it substantially faster and suitable for real-time applications. Its simplicity and effectiveness have made YOLO a popular or even the go-to choice among researchers and developers in the field, spurring continuous development and improvements across its subsequent versions. From the original YOLO [32] to YOLOv8 [23], each iteration brings improvements in speed and accuracy, and we also see different versions optimized for hardware platforms of varying computing power [34].

For visually impaired users, the application of YOLO-based object recognition systems has the potential to dramatically improve spatial awareness and personal safety. This is primarily due to YOLO’s capability to recognize objects in real time. Such rapid processing is invaluable for safe interaction and navigation within various environments. Studies exploring YOLO’s integration into assistive technologies have consistently highlighted its effectiveness in detecting and classifying objects under diverse conditions, significantly aiding visually impaired individuals’ navigation capabilities [37].

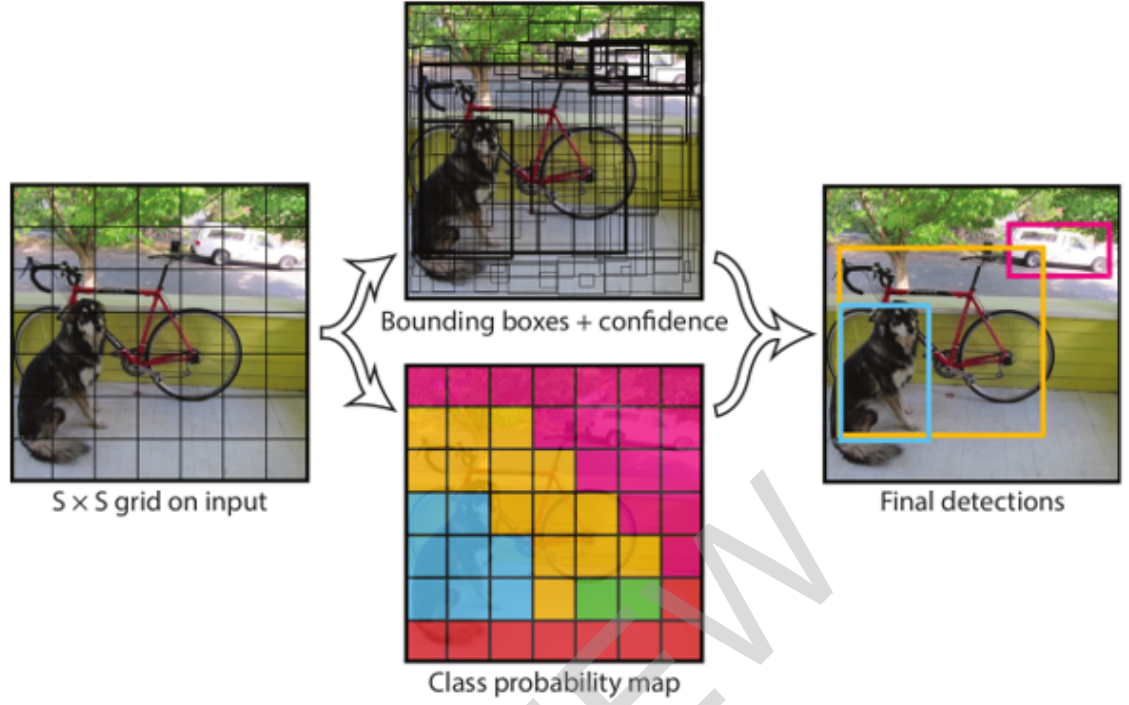


Figure 2.1. YOLO Object Recognition Procedures [32]

Many different object recognition systems utilizing YOLO have been developed for the visually impaired in the past few years. In 2018, one of the first such systems was developed to detect traffic lights [38]. Running on an NVIDIA GTX 1080 Ti GPU, the system can reach up to 250 Frames Per Second (FPS) using YOLO Tiny, and 100 FPS using YOLO Full. The system also showed that YOLO is substantially faster than Faster R-CNN (0.37 FPS) and SSD (50 FPS). In 2020, [39] utilized YOLOv3-tiny to detect 80 classes of items, with both Android application and web application. In 2021, [40] obtained 12 FPS using NVIDIA GTX 1050 GPU, but the frame rate dropped to 2 using NVIDIA Jetson Nano. In both configurations, YOLOv3 was used. The author also used monocular depth estimation for distance measurement. One year later, [41] utilized YOLOv4-tiny running on a Raspberry Pi 4B for object recognition with depth estimation. However, the frame rate of the system was not mentioned. Mobile apps were also developed with YOLO for object detection [42].

From the review of YOLO-based object recognition systems listed above, we can see a trend that incorporates YOLO into portable assistive devices, making it feasible to implement this powerful object recognition tool in wearables and mobile applications. However, while the evolution of the YOLO algorithm from the original YOLO to YOLOv8 has seen improvements in detection precision, processing speed, and the algorithm's robustness against varying conditions, we did not see enough research using more advanced versions of YOLO to improve performance. We aim to address this deficiency in our research.

2.2 Obstacle Avoidance Systems

Obstacle avoidance is a key functionality for almost all systems designed to assist visually impaired individuals. These systems, leveraging a variety of sensing technologies and computational algorithms, identify potential hazards in the users' path and alert them through different feedback mechanisms. The evolution of obstacle avoidance strategies has been closely tied to advancements in sensor technologies, including ultrasonic sensors, LIDAR, and depth-sensing cameras, each offering unique benefits and challenges in accurately detecting obstacles in real time [26]. Also, in terms of algorithms used, various approaches have been developed to tackle this challenge, ranging from traditional sensor-based feedback loops to more advanced machine learning-based solutions [27].

In terms of obstacle avoidance technologies, they can be broadly categorized into non-vision-based and vision-based systems, each leveraging different mechanisms to detect and navigate around obstacles. The former consists of ultrasonic sensors, infrared sensors, Lidar (Light Detection and Ranging) sensors, and mmWave radar, while the latter consists of single cameras, stereo (dual) cameras, and RGB-D cameras. Should a system utilize technologies from both groups, we can call it a hybrid system. This is shown in Figure 2.2. We shall discuss examples in subsections below.

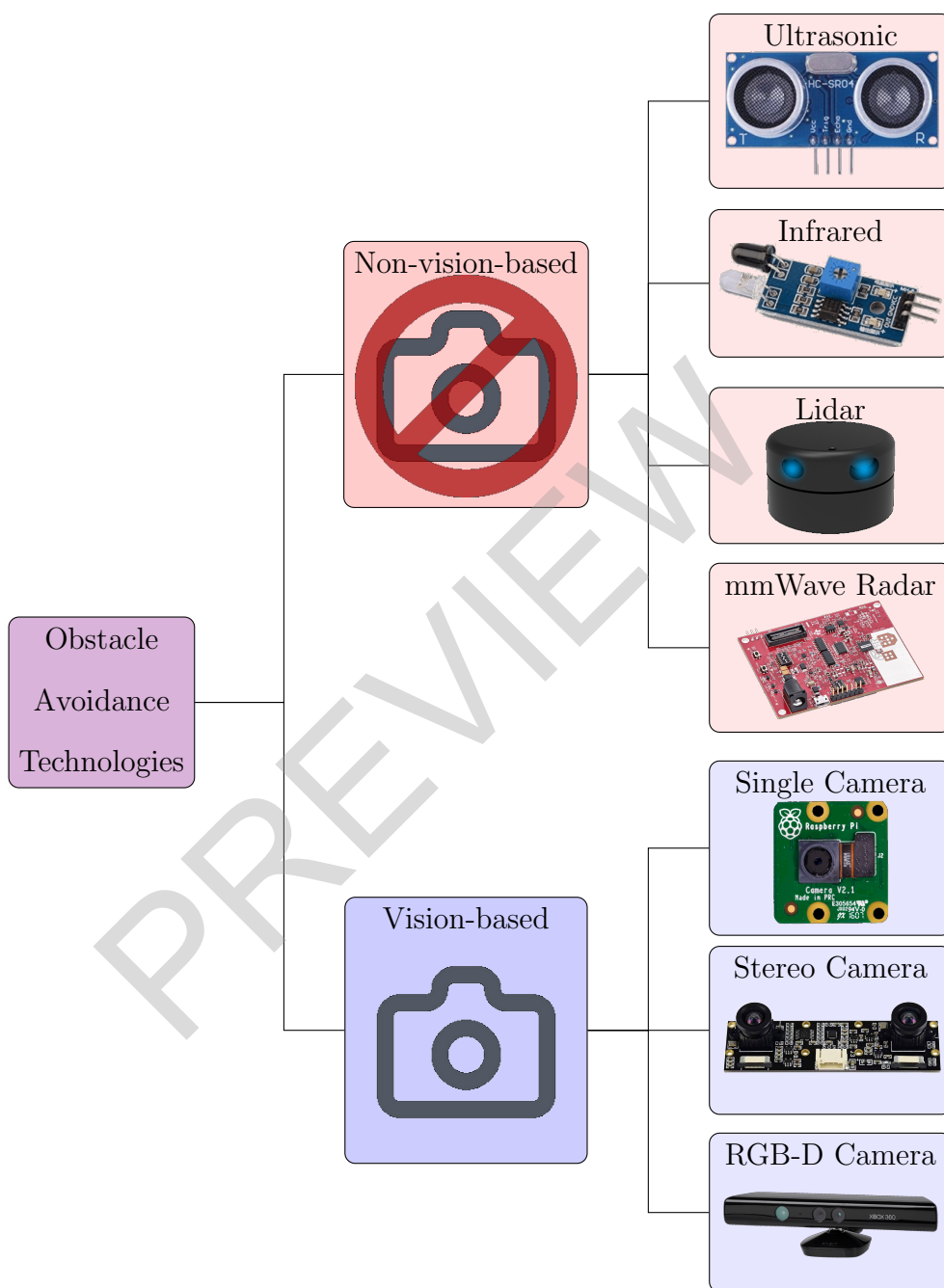


Figure 2.2. Obstacle Avoidance Technologies

2.2.1 Non-Vision-Based Obstacle Avoidance Technologies. Non-vision-based obstacle avoidance technologies rely on sensors that do not require visual input to detect obstacles. These systems are particularly advantageous in environments where lighting conditions or visual obstructions may hinder the efficacy of vision-based systems. However, they come with inherent limitations, a primary one being their inability to recognize the type of obstacles they detect [27]. This limitation stems from the nature of the sensory data they collect, which, unlike visual data, often lacks the details necessary to distinguish between different objects or environmental features. We delve into different non-vision-based technologies below. A comparison of these technologies in terms of operation principle, range, accuracy, cost, special ability, and obstacle type recognition ability is provided in Table 2.1.

Table 2.1. Comparison of Non-vision-based Technologies for Obstacle Avoidance

Sensor Type	Ultrasonic	Infrared	Lidar	mmWave Radar
Principle of Operation	Time-of-Flight (ToF)	Return Signal Intensity	ToF	ToF
Typical Range	A few meters	A few meters	Up to 200 meters Close distance limited	>200 meters Close distance limited
Accuracy	Moderate	Moderate	High	High
Cost	Low	Low to Moderate	Moderate to High	Moderate to High
Special Ability	Limited	Limited	360-degree coverage, accurate mapping	Excellent penetration through fog, smoke, and even thin materials
Obstacle Type Recognition Ability	None (distance only)	None (distance only)	Low (detects outlines at best)	Moderate (some capable of low-resolution imaging)

2.2.1.1 Ultrasonic Sensors. Ultrasonic sensors are among the most widely adopted technologies for obstacle detection. They operate by emitting ultrasonic sound waves that bounce off objects and return to the sensor. By measuring the time it takes for these sound waves to return, the sensors can calculate the distance to obstacles. This technology is prized for its simplicity and cost-effectiveness. Ultrasonic sensors can reliably detect objects within a short to medium range with good accuracy [27], making them suitable for indoor navigation and close-quarters maneuvering. In Figure 2.3, we have an illustration showing the working principle of ultrasonic distance sensors.

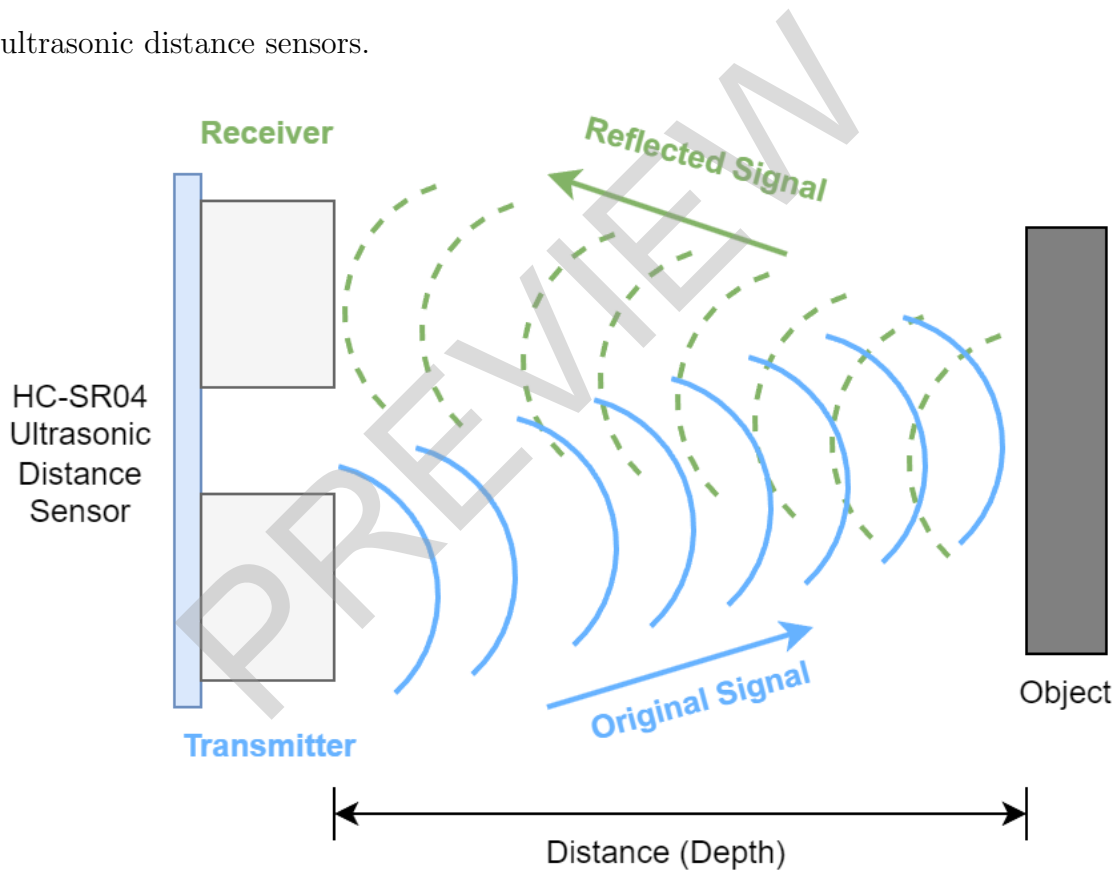


Figure 2.3. Working Principle of Ultrasonic Distance Sensor

A basic obstacle avoidance system using ultrasonic sensors alone typically needs only the ultrasonic sensor and an embedded system such as Arduino. An example is the system developed in [43], which uses one HR-SR04 ultrasonic sensor

and an Arduino Nano as the processing unit. The range of this system is 4 meters. Such systems have a low cost but lack object recognition capabilities. By using multiple sensors with better performance, it is possible to improve the obstacle detection range and field of view [44]. However, it is still not possible to address the problem of object recognition with only ultrasonic sensors. Two other problems faced by the ultrasonic approach are the inability to detect objects with smooth reflective surfaces, and cross-talk among multiple sensors [27].

2.2.1.2 Infrared Sensors. Similar to ultrasonic sensors, Infrared (IR) sensors detect obstacles by emitting infrared light and detecting the light reflected from surfaces. These sensors are highly effective for short-range detection and are known for their precision and low power consumption. IR sensors are particularly useful in controlled environments where the reflective properties of objects and surfaces can be anticipated. However, their performance can vary significantly with environmental conditions, such as the presence of sunlight or other infrared sources, which can interfere with detection accuracy. In the presence of powerful natural light such as sunlight, infrared sensors may be unable to obtain correct measurements [45].

Because of their inherent limitations, infrared sensors are not as widely used as ultrasonic sensors in the field of obstacle avoidance systems. While low-cost infrared sensors are available, they mostly offer a shorter range compared with ultrasonic sensors in the same price range. One example is [46], which utilized one E18-D80NK infrared sensor, and has a maximum range of 0.8 meters. On the other hand, high-end infrared sensors exist, and they offer a better range and update rate than most ultrasonic sensors. A notable example is given in [47], whose system has a maximum range of 14 meters. Also, by using a total of 7 TeraRanger One sensors, the system has a wide field of view, with the capability of detecting obstacles in front and next to the feet of the user. However, even high-end infrared sensors share the same problem with